GeaNaVid

Gea, Na or Vid: verifying claims against Swiss federal voting booklets

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Team Apertool · Hack Apertus 2026 · Track 2A — OST: Multilingual NLI over Swiss Voting Booklets

- **Track:** Track 2A — OST: Multilingual NLI over Swiss Official Voting Booklets
- **Event:** Online
- **Team:** Apertool — Niklas Wichter
- **Demo:** [TBD: video link] · `make run` in `track_2a/`

Draft of 4 October 2026. All numbers are from runs on OST v1.0 stored in `runs/v1/`; open measurements are marked [TBD]. Ablations, retrieval recall, calibration, error analysis and the full tables are in `docs/EXPERIMENTS-v1.md`; the improvement rounds, including the rejected ones, in `docs/ITERATIONS.md`.

1. Summary

Swiss voters meet many claims about federal proposals; the official voting booklet is the authoritative source, but it runs to 40–90 pages in three languages. GeaNaVid checks a claim in German, French or Italian against a booklet and returns entailment, neutral or contradiction, for entailment and contradiction together with the passages that decide the label, plus the token usage and the time. We treat the task as document-level natural language inference (Koreeda & Manning, 2021; Yin et al., 2021), close to claim verification with evidence (Thorne et al., 2018), but judged only against the booklet, never against the world.

Our CLI `booklint` parses the PDF without a language model, narrows the booklet to the proposal named in the case's `vote` field, retrieves a few passages with BM25, preferring the section the claim names ("the summary", "the committee", "the voting text"), and asks Apertus v1.5 for a strict JSON verdict with evidence first and label second. Cross-lingual claims are translated into the booklet language by Apertus before retrieval; with 8B, a neutral answer is checked once more, only for a contradiction. On the official OST v1.0 benchmark (497 held-out test pairs), Apertus 8B reaches a macro-F1 of 0.957 [95% CI 0.933–0.978, cluster bootstrap] with 2,662 input tokens per claim, against 0.723 for the whole-booklet baseline (20,156 input tokens). Apertus 70B reaches 0.968 [0.948–0.984] with 1,531 input tokens, against 0.890. The evidence sentence lies in OST's gold section for 86 % (8B) and 88 % (70B) of the answers. In the Beginner task (gold reference given), 8B reaches 0.978 and 70B 0.980. On a synthetic hold-out from booklets outside OST v1.0 the gap to the baseline carries over (8B 0.874 vs 0.665, 70B 0.932 vs 0.872). The output is never a voting recommendation: every yes or no comes with a quote from the booklet; a neutral answer ("blank") cites nothing.

2. System design and architecture

The CLI reads OST's case format (`id`, `booklet.path` or `reference.text`, `vote`, `claim.text`) and prints OST's output format (`id`, `label`, `label_name`, `evidence[{page, text}]`, `metrics{input_tokens, output_tokens, inference_time_ms}`). The default pipeline for the Advanced task:

1. **Ingest** (once per booklet, no LLM, cached): text blocks with page, proposal, section and committee note.
2. **Languages** of claim and booklet from function words and letters; no language field needed.
3. **Proposal**: the booklet is narrowed to the proposal named by `vote` (BM25 over proposal titles and texts). A `vote` title in another language than the booklet is first translated by Apertus.
4. **Translation**: a cross-lingual claim is translated into the booklet language by Apertus.
5. **Retrieval**: BM25 top-5 over claim and translation inside the proposal, plus same-page neighbours; blocks of the section the claim names (summary, committee, voting text, Federal Council) rank higher.
6. **Apertus NLI**, one call: numbered passages, JSON schema with evidence before label. With 8B the prompt carries three development examples, a verdict with p(label) < 0.9 is re-checked by splitting the claim into atomic statements (two more calls), and a neutral verdict is rechecked once with a narrow question: does a passage contradict one detail of the claim (number, actor, position, direction, scope)? For a cross-lingual committee claim, "Ausschuss" / "commission" in the translation is set back to the booklet's "Komitee" / "comité" / "comitato".
7. **Evidence sentence** (no LLM): the sentence of the cited or retrieved passages, or of the named section, that covers most of the claim, with its page.

For the Beginner task (claim + reference text) steps 3 and 5 are skipped: the reference is the context. A failed case (missing or scanned PDF, endpoint error) is answered neutral with an `error` field, so a batch never stops.

2.1 Document parsing and context preparation

Parsing. PyMuPDF text blocks are cleaned (soft hyphens, line-break hyphenation, footnote digits that otherwise glue to numbers). Each block keeps its page (1-based PDF index), the proposal from the running header, the nearest bold side heading ("In Kürze", "Empfehlung von Bundesrat und Parlament") and the committee note. Parsing is deterministic and costs no tokens. Booklets without a text layer are reported as failed cases (no OCR yet).

Context preparation. (a) Locating the proposal by the `vote` title keeps 93.7 % of the gold section inside about a quarter of the booklet (median 19 k of 82 k characters). (b) BM25 (Robertson & Zaragoza, 2009) over heading plus text, with number normalisation and prefix stemming, fused across claim and translation by reciprocal rank (Cormack et al., 2009); BM25 needs no GPU and is a strong zero-shot baseline (Thakur et al., 2021). (c) **Sections.** Every block gets a section type from its headings and page notes (legal text,

committee, Federal Council arguments, recommendation, explanation; DE/FR/IT keywords), and the claim's wording tells which source it names. On the 998 development claims the gold section contains the explanation part for 99 % of "summary" claims, the committee pages for 93 % of committee claims and the recommendation or Federal Council arguments for 98 % of Federal Council claims. Blocks of the named section get a factor 1.5 in the rank fusion. hit@5 of the gold section is 0.980 on test with translation and sections (0.928 without sections, 0.718 without translation, 0.604 over the whole booklet). (d) The five hits and their same-page neighbours, each prefixed with page, proposal and section, give about 1.1 k tokens of context instead of 20 k for the whole booklet (p95 30 k).

NLI call and evidence. Apertus names the deciding passage numbers before the label, as in evidence-first setups (Koreeda & Manning, 2021; Thorne et al., 2018). The returned evidence is a verbatim booklet sentence chosen after the decision without an LLM call (idf-weighted word stems and character 5-grams against claim and translation; sentences of the named section count double, and its four best blocks are candidates too); `--evidence-unit paragraph` returns the whole text block.

Not in the default. A page-window ladder (Li et al., 2024), an 8B → 70B cascade (Chen et al., 2023), the parallel language version instead of translation, dense or hybrid retrieval, Apertus reranking, label calibration, number rules, compact passage headers, fewer passages without neighbours, re-checking answers with $p < 0.999$, re-checking entailments for a contradiction, translating the passages instead of the claim and a sharper contradiction/neutral definition in the prompt are flags or were dropped; on development data none paid for its cost, and the neutral re-check did not pay on test for 70B (`docs/EXPERIMENTS-v1.md`, `docs/ITERATIONS.md`).

3. Use of Apertus

- **Model:** swiss-ai/Apertus-v1.5-8B and swiss-ai/Apertus-v1.5-70B (Project Apertus et al., 2025)
- **How it is used:** inference only: the NLI verdict with evidence selection, the translation of cross-lingual claims and of foreign `vote` titles, and (8B) the split of unsure claims into atomic statements and the contradiction re-check of neutral verdicts.
- **Where it runs:** the Hack Apertus OpenAI-compatible endpoint, selected through `LLM_NAME`, `LLM_BASE_URL`, `LLM_API_KEY`; temperature 0, JSON schema, `logprobs` with 10 alternatives.

The label probability is read from the first token of the label value. No other model takes part in the default path. Other models only supported development: Claude (Anthropic) wrote the claims of the synthetic hold-out, and Codex CLI (GPT-6.1-sol) and NVIDIA Nemotron 3.5 Lightning (open weights) checked its labels as blind judges; none of them runs in the pipeline or produces a reported OST number. Every call is cached on disk by a hash of model, messages and parameters, which makes reruns free and reproducible.

4. Evaluation

Data. OST v1.0 (MIT) has 1,495 claim-booklet pairs, 60 booklets from 20 voting dates (2020–2026), 177 proposals, balanced labels and 974 pairs (65 %) cross-lingual (`OSTswiss/MNLIoverSwissVotingBooklets`). Split by voting date (seed 2026), so the DE/FR/IT versions of a booklet never cross splits: 998 development pairs (13 dates), of which a stratified subset of 450 was used for all ablations, and 497 test pairs (7 dates, 21 booklets). Variants were developed on `dev_s`; a default changed only if it was better on `dev_s` and test (five test evaluations, `docs/ITERATIONS.md`), so test is not fully untouched and the synthetic hold-out below (0.874 / 0.932) is the independent check. v1.0 contains 335 rows that are exact copies of another row (same claim, booklet and label), 122 of them in test (375 distinct pairs). We keep them as published; a plain bootstrap then counts copies as independent evidence and its intervals are too narrow. Recomputed from the stored predictions (no new calls), the defaults score 0.955 (8B) and 0.965 (70B) without copies; resampling each group of copies as one unit gives the intervals [0.933–0.978] and [0.948–0.984] (by proposal: about the same). All reported numbers use OST v1.0 only. Two auxiliary sets written by closed LLMs are used for no reported number: 24 unreviewed edge cases (Gemini 3.8 Flash via the Antigravity CLI), demo presets only, and never evaluated because 16 of them concern proposals of the test dates; and archived synthetic development sets from before v1.0 (Claude Opus 5.5, GPT-5.5). A unit test checks that no stored run contains an edge case; sources and per-row provenance are in `data/README.md`.

Metrics as on OST's slide 28: macro-F1 over the three labels with 95 % bootstrap intervals (2,000 resamples, each group of copies as one unit, `runs/v1/test/duplicates-ci.json`), separately for Beginner and Advanced and by language; evidence against the gold section (`reference_string`) and its page; input tokens per claim (what OST sums); mean and p95 of `inference_time_ms`.

Main results (test, macro-F1; input tokens per claim, 8B / 70B where they differ):

Setup	8B	70B	Input tok.
Baseline: whole booklet	0.723	0.890	20,156
Beginner: gold section	0.940	**0.980**	2,080
Beginner + 3 examples (8B)	**0.978**	–	2,662
Adv.: BM25, whole booklet	0.762	–	1,367
Adv.: proposal + BM25	0.778	0.819	1,312
Adv.: + translation	0.940	**0.951**	1,636 / 1,631
Adv.: + 3 ex. + decomp.	0.941	–	2,258
Adv.: + sections (70B default)	0.953	**0.968**	2,129 / 1,531
Adv.: + neutral re-check (8B default)	**0.957**	0.966	2,662 / 2,066

Defaults: 8B the last row (three examples, decomposition, neutral re-check), 70B the row before. The 8B pipeline beats the 8B baseline by +0.234 and even the 70B baseline; 70B gains +0.078. Both need 1.5–2.7 k instead of 20 k input tokens per claim. Each step was developed on the 450 development claims and kept only if test agreed; the section step added +0.011 (8B) and +0.015 (70B) on development data and +0.013 / +0.017 on test; the neutral re-check with the committee words added +0.016 / +0.009 on development data and +0.004 / −0.002 on test, so 70B does not use it (a decision that used test).

By language (test, macro-F1; claim or booklet language, n in brackets):

Group (n)	Adv. 8B	Adv. 70B	Beg. 8B	Beg. 70B
All (497)	0.957	0.968	0.978	0.980
Claim DE (166)	0.951	0.964	0.956	0.963
Claim FR (165)	0.945	0.958	0.982	0.989
Claim IT (166)	0.976	0.981	0.994	0.988
Booklet DE (156)	0.981	0.962	0.974	0.987
Booklet FR (186)	0.966	0.983	0.983	0.989
Booklet IT (155)	0.920	0.951	0.974	0.965
Same lang. (166)	0.987	0.987	0.980	0.987
Cross-ling. (331)	0.943	0.958	0.976	0.977

The whole-booklet baseline drops on cross-lingual pairs (8B 0.801 same vs. 0.685 cross); translation closes most of the gap. With the claim language detected instead of read from the data, the Advanced results are identical (detection right on 497 of 497 test claims).

Synthetic hold-out. To check that our choices did not fit OST's 20 voting dates, we built 89 claims on five proposals of three voting dates outside v1.0 (2018–2019, older layouts; written by Claude, labels confirmed blindly by Codex and Nemotron, one disagreement dropped; not human-annotated, never mixed with OST numbers) and evaluated them once after all defaults were fixed: whole booklet 0.665 (8B) / 0.872 (70B), GeaNaVid 0.845 / 0.932 at 9 % / 7 % of the tokens, Beginner 0.855 / 0.931 (`data/holdout-synth/`). The 8B neutral re-check, added later, was evaluated there once more: 0.874 (contradiction F1 0.735 → 0.793).

Evidence (Advanced). The first evidence sentence lies inside the gold section for 0.861 (8B) and 0.881 (70B) of entailment/contradiction answers, and on a gold page for 0.870 / 0.890; one of the (up to two) sentences does for 0.947 / 0.947. Before the section step these were 0.786 / 0.804: the booklet often says the same thing in the summary, the details and the legal text, and the claim's wording tells which one is gold. In a read sample of 30 sentences (8B, before the section step), 22 decide the claim, 7 partly, 1 does not. How OST matches evidence is not yet known, so OST's evidence score is [TBD].

Efficiency as OST measures it. Cold run on 50 seeded test cases in OST's input format, no response cache, one request at a time; `inference_time_ms` is the wall time of one case (booklet parsed beforehand):

Model	Input tok.	Calls	Mean ms	p95 ms
8B	2,761	2.10	648	1,188
70B	1,564	1.68	1,344	2,200

The 50 labels equal those of the stored runs for both models. The neutral re-check costs 8B about 250 ms on average (the previous 8B default measured 401 / 709 ms in the same session). Our own code takes 48 ms per claim (parsing is cached per booklet); the rest is Apertus. The endpoint's prefix cache cannot be switched off and its speed varies (the previous 8B defaults measured 457 ms in an earlier cold run); tokens counted by OST's proxy are [TBD].

5. Reproducibility

`make run` builds the Docker image and runs the default pipeline on the test split against the endpoint in `LLM_*`; no GPU or local weights are needed; from the response cache it reproduces 0.957 (8B) and 0.968 (70B). For OST's own cases: `booklint predict --case <cases.jsonl>` (mounts and `BOOKLET_ROOT` in `track_2a/README.md`). Settings: prompt `nli-v1`, k = 5, one neighbour, temperature 0, JSON schema. Replaying the stored test runs from the response cache with the current code gives the same 497 labels for both models; each run in `runs/v1/` stores its options, split seed and command. At temperature 0 the endpoint is not fully deterministic: a cold rerun of 50 cases gave the same label on 50/50 (8B) and 49/50 (70B). Commit of the submission: [TBD].

6. Limitations

497 test claims (375 distinct) from 7 voting dates: differences below about 0.02–0.03 macro-F1 are within the cluster-bootstrap interval (70B − 8B: +0.010 [−0.012; +0.035]), and the model-aware defaults (examples, decomposition and neutral re-check only for 8B) were chosen on 450 development claims. The section keywords cover the federal booklet layouts of 2018–2026; other layouts fall back to plain BM25. Main error sources are topical but unsupported claims, attribution (committee vs. Federal Council) and contradictions that change one detail, which Apertus calls neutral with high confidence (no probability gate finds them; the narrow re-check finds some for 8B, while 70B then over-calls contradiction). Booklets without a text layer are not handled (OST provides text PDFs). Labels and evidence are an automated reading of the booklet, not a voting recommendation.

Open questions to OST (Q&A on 2026-10-08): whether `page` is the PDF index or the printed page number; which page the Beginner task expects (we return `page: 1`); how evidence is matched; and whether the token-counting proxy passes `response_format` and `logprobs` through. Without them the client retries without schema and log-probabilities and the 8B decomposition is skipped (about −0.004 macro-F1 on development data).

7. Lessons learned

- **Narrow first, then read.** Locating the proposal by its title costs nothing; with translation it beats the whole booklet by +0.22 (8B) and +0.06 (70B) with 1.6–2.3 k instead of 20 k input tokens. 8B with this context (0.941) beats 70B with the whole booklet (0.890).
- **Translation is the largest single step** for cross-lingual claims (8B cross-lingual 0.696 → 0.938).
- **Prompt rules move the boundary instead of sharpening it.** Rules for "neutral" made 8B call real contradic-

tions neutral; three worked examples helped on development data but not on test for the Advanced task. Apertus is very confident (p > 0.9 for 94–98 % of answers), so probability gates rarely fire and calibration offsets were unstable.

- **A narrower second question beats a gate.** The neutral answers that hide a contradiction have p ≈ 1, but asking only "is one detail contradicted?" turns 5 of 11 such 8B errors on development data into right answers without a single wrong flip. The same question after an entailment flags most correct answers.
- **Cheap evidence.** Choosing the evidence sentence after the decision costs no tokens and lands in the gold section more often than the whole cited passage (0.79 vs. 0.62 for 8B).
- **Read where the claim says it comes from.** "According to the summary" or "the committee thinks" names the section that holds the answer; routing on these words raised evidence hits by 7–8 points and macro-F1 by 0.013–0.017 at fewer tokens, with no model call.
- **Measure what the judges measure.** OST sums input tokens and takes mean and p95 of time per claim; our first numbers (total tokens, warm cache) had to be re-measured cold.
- **Check the input format, not only the dataset.** OST's cases have no language field and may carry a `vote` in another language; translating the title raises the right proposal from 70 % to 95–98 %.

Next steps: settle evidence unit and page numbering with OST, extend to Romansh (Stüssi & Vamvas, 2026).

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References

Chen, L., Zaharia, M., & Zou, J. (2023). FrugalGPT: How to Use Large Language Models While Reducing Cost and Improving Performance. *Arxiv Preprint Arxiv:2305.05176*. https://arxiv.org/abs/2305.05176

Cormack, G. V., Clarke, C. L. A., & Buettcher, S. (2009). Reciprocal Rank Fusion Outperforms Condorcet and Individual Rank Learning Methods. *Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, 758–759. https://doi.org/10.1145/1571941.1572114

Koreeda, Y., & Manning, C. (2021). ContractNLI: A Dataset for Document-level Natural Language Inference for Contracts. *Findings of the Association for Computational Linguistics: EMNLP 2021*, 1907–1919. https://doi.org/10.18653/v1/2021.findings-emnlp.164

Li, Z., Li, C., Zhang, M., Mei, Q., & Bendersky, M. (2024). Retrieval Augmented Generation or Long-Context LLMs? A Comprehensive Study and Hybrid Approach. *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing: Industry Track*, 881–893. https://doi.org/10.18653/v1/2024.emnlp-industry.66

Project Apertus, Hernández-Cano, A., Hägele, A., & others. (2025). Apertus: Democratizing Open and Compliant LLMs for Global Language Environments. *Arxiv Preprint Arxiv:2509.14233*. https://arxiv.org/abs/2509.14233

Robertson, S., & Zaragoza, H. (2009). The Probabilistic Relevance Framework: BM25 and Beyond. *Foundations and Trends in Information Retrieval*, *3*, 333–389. https://doi.org/10.1561/1500000019

Stüssi, E., & Vamvas, J. (2026). How Good is AI on Swiss Voting Booklets? A Multilingual OCR and Alignment Benchmark. *Proceedings of the 11th Edition of the Swiss Text Analytics Conference*, 232–242. https://aclanthology.org/2026.swisstext-1.22/

Thakur, N., Reimers, N., Rücklé, A., Srivastava, A., & Gurevych, I. (2021,). BEIR: A Heterogenous Benchmark for Zero-shot Evaluation of Information Retrieval Models. *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*.

Thorne, J., Vlachos, A., Christodoulopoulos, C., & Mittal, A. (2018). FEVER: a Large-scale Dataset for Fact Extraction and VERification. *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long Papers)*, 809–819. https://doi.org/10.18653/v1/N18-1074

Yin, W., Radev, D., & Xiong, C. (2021). DocNLI: A Large-scale Dataset for Document-level Natural Language Inference. *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021*, 4913–4922. https://doi.org/10.18653/v1/2021.findings-acl.435